



The Global Military-Industrial Base: A Predictive and Diagnostic Analysis

Collaborators

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Introduction

In an era defined by geopolitical uncertainty, technological advancement, and shifting global alliances, military expenditure has become a critical indicator of national priorities and industrial strength. Beyond defense of readiness, military spending reflects broader economic capabilities, innovation pipelines, and the scale of a nation's industrial base.

This project presents a comprehensive analytical exploration of the global military-industrial landscape using Power BI. By integrating multiple datasets on national defense spending and leading defense manufacturers, the dashboard aims to uncover patterns that are not immediately visible through isolated data sources. The objective is to move beyond static reporting and develop an interactive, insight-driven platform that supports both diagnostic and predictive analysis.

Through this work, we seek to answer a central question which nations are emerging as dominant industrial and defense powerhouses and what trends in spending



and production support this transformation? By combining financial, industrial, and temporal dimensions, the project provides a nuanced perspective on how global defense ecosystems are evolving.

Data Sources and Context

This project leverages three primary datasets that collectively provide a multi-dimensional view of the military-industrial domain:

Datasets:

- Global Military Spending Dataset
- Top Defense Manufacturers Dataset
- SIPRI Military Expenditure Database

These datasets were selected for their complementary nature. While the SIPRI and global spending datasets provide country-level insights into defense budgets over time, the defense manufacturers' dataset introduces an industrial perspective by highlighting key players contributing to military production.

By merging these sources, the analysis bridges the gap between national expenditure and industrial output, enabling a more complete understanding of how financial investments translate into defense capabilities and economic influence.



The combined dataset allows for:

- Cross-country comparisons of military spending trends
- Identification of high-growth regions and emerging powers
- Correlation between national budgets and manufacturer revenues
- Exploration of long-term global defense patterns

The overarching goal is to construct a predictive and diagnostic view of the global military-industrial base, enabling users to identify trends, anomalies, and strategic shifts.

Datasets:

<https://www.kaggle.com/datasets/konradb/global-military-spending-dataset>

<https://www.kaggle.com/datasets/surajjha101/top-defense-manufacturers>

<https://www.kaggle.com/datasets/azeeshan20/sipri-military-expenditure-database>

These are related datasets on military spending and top defense manufacturers, and when combined, it will help us with a better story telling experience. The primary goal is to provide a predictive and diagnostic view of the global military industrial base. By merging SIPRI/GMSD spending data with top defense manufacturer revenue, we aim to answer: Which nations are becoming industrial powerhouses and determine identifiable trends in spending and explore helpful insights that could be determined from this analysis.



Data Cleaning and Preparation

The datasets required substantial standardization before analysis could begin, particularly at the country level. Because the sources used different naming conventions, formatting styles, and historical labels, country values were cleaned, reordered where necessary, capitalized consistently, and manually corrected for selected cases to ensure

alignment across tables. Text fields were trimmed, blank values were converted to nulls where appropriate, and invalid currency entries were removed before spending values were converted into a fixed decimal format. These transformations ensured that the data could be accurately compared, aggregated, and visualized in Power BI.

- **String Manipulation for Country Normalization:** We developed a custom function to handle reversed names (e.g., "Korea, South" to "South Korea") and used "Capitalize Each Word" to ensure case-sensitive joins remained consistent.
- **The Country Mapping:** Because different datasets use varied naming conventions (e.g., "USA" vs. "United States of America"), we utilized mapping table that links all data points to a universal Country Code.
- **Financial Sanitization:** We filtered out non-numeric values such as "xxx" or "." from the SIPRI Value column and converted the data into a **Fixed Decimal (\$)** format to ensure mathematical precision in revenue and spending calculations.
- **Handling Nulls and Datatypes:** To prevent errors in Power BI's DAX engine, we systematically removed nulls from currency columns and applied "Trim" and "Clean" functions to all text-based fields.



Data Modeling:

The project transitioned from flat-file analysis to a sophisticated Star Schema.

- **Central Value Anchor:** The model is centered around a unified Country code dimension table, which serves as the "source of truth" connecting all disparate data points.
- **Fact Table Integration:** Sovereign spending (SIPRI), GDP-related indices (Milex), and corporate rankings (Top Manufacturers) are linked to the central country dimension via many-to-one relationships.
- **Multi-Dimensional Connectivity:** The model bridges historical expenditure tables with real-time industrial revenue, allowing for cross-table filtering between economic performance and defense output.
- Though the star schema gave high-level insights into the defense data, more sophisticated analysis needed complex joins, so new relationships had been defined to manage the complexity.

Designing the Dashboard

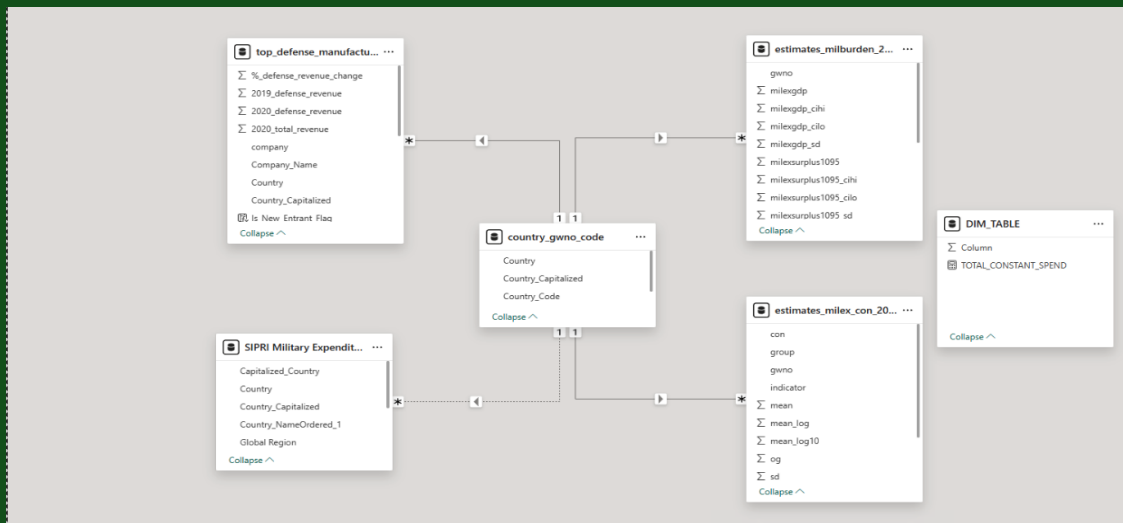
- **Geo-Spatial Intelligence:** An interactive map uses bubble markers to visualize global spending concentrations, immediately identifying regional powerhouses.
- **Industrial Matrix:** A heat-map matrix cross-references specific manufacturers against countries to detect "industrial insourcing" and domestic dominance.

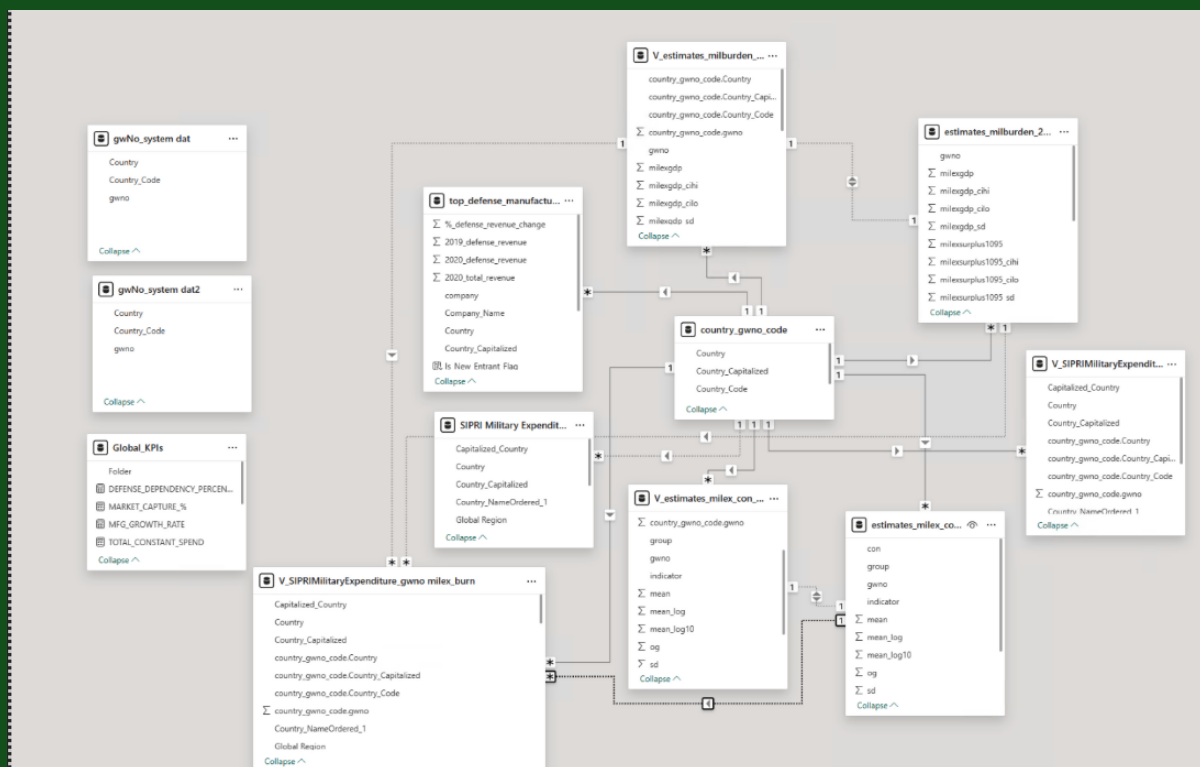


- **Temporal Trend Analysis:** As shown stacked area charts and donut visualizations track the "Sum of Standardized Spend" by year and global region, providing a longitudinal view of how regions like Europe or Asia & Oceania have shifted their fiscal priorities over decades.
- **Interactive Slicers:** Users can filter by specific years or global regions to dynamically update the entire analytical view.



Star Schema





Final Data Model

Telling the Story Through Analysis

Our analysis goes beyond basic reporting to provide diagnostic insights by comparing a nation's total spending with the revenue of its domestic manufacturers we can identify which countries are becoming self-sufficient industrial hubs. The dashboard highlights



Defense Dependency, allowing us to audit which nation's economies are most heavily tied to the military sector. Spikes in the Total Spending line often precede shifts in manufacturer rankings, providing a leading indicator for future industrial growth.

Design Choices and Visualization Strategy

The dashboard uses a consistent orange and green color palette across all pages, making it easy to distinguish between regions and companies at a glance. Categorical colors are assigned per Global Region of Africa, Americas, Asia & Oceania, Europe, and Middle East and remain consistent across every visual, so a viewer never has to re-learn the legend on a different page.

Chart types were chosen to match the specific analytical question for each visual answer. Bar charts are used for country-level comparisons because they make ranking immediately obvious as the US bar's length alone communicates its dominance without any annotation. The stacked area chart shows both total growth and regional composition shifting over time simultaneously, something a simple line chart cannot do. The donut chart breaks down sub-regional share cleanly, with the Americas 76.4% slice immediately visible. The matrix handles the multi-country, multi-company comparison where a bar chart would become too cluttered, and its conditional color formatting green to amber to orange encodes spending magnitude, so viewers can rank values without reading every number.

Interactivity was designed to minimize navigation friction. Every visual use cross-filtering, so clicking a country or region in one visual instantly updates all others on the same page. The year slicer and Global Region filter allow users to step through time by period and

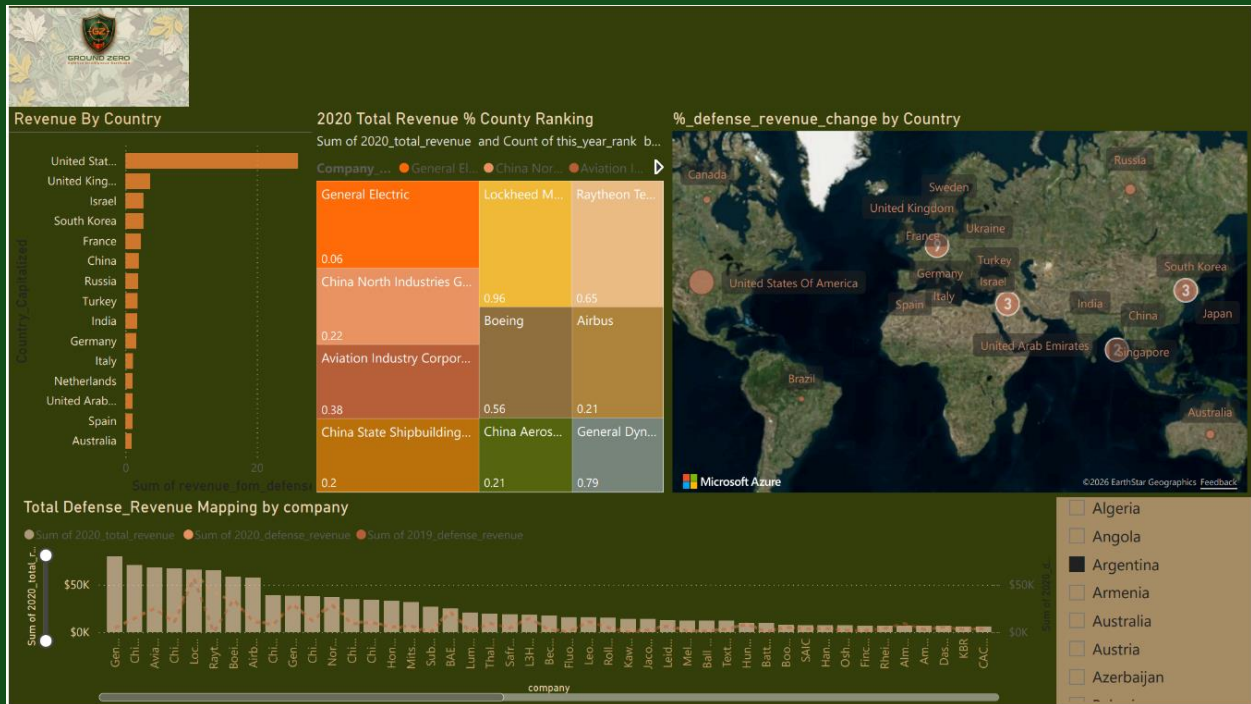


watch the regional composition of the donut and area chart update in real time, turning what could be a static snapshot into a genuinely exploratory analytical tool.

Dashboard References

Global Defense Industrial Map

- This view maps the commercial side of defense. It identifies which nations and specific companies are capturing the most value from global military expenditures.
- **This** Highlight the dominance of the United States, China, and the United Kingdom in total defense revenue.
- **Market Leaders:** Tree map to show how massive firms like Lockheed Martin, Boeing, and China North Industries Group define the industrial landscape.
- **Geopolitical Footprint:** The map visualizes the hubs of production showing where industrial power is physically concentrated.

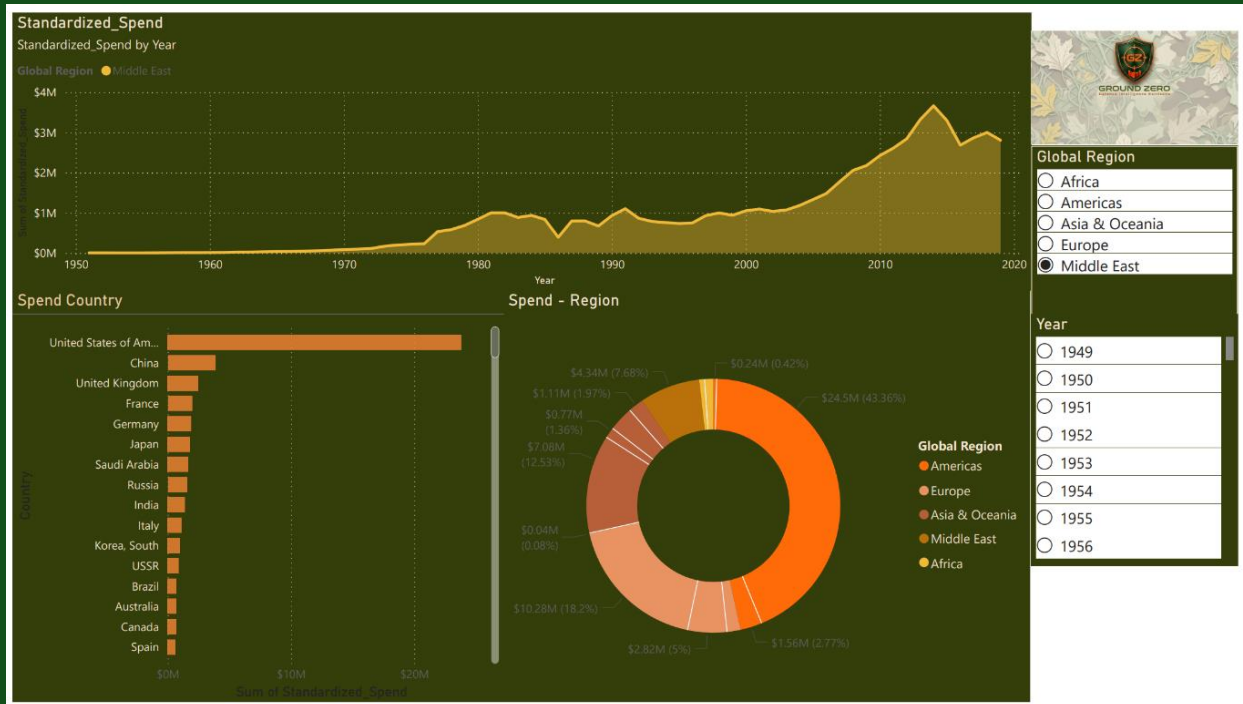




2. Longitudinal Expenditure Trends

Historical Velocity & Regional Dominance

- This dashboard tracks Standardized Spend over 70 years identifying historical peaks and regional shifts.
- **The 2010 Peak:** Highlight the significant global rise in spending leading up to 2010, followed by the relative stabilization seen in the late 2010s.
- **Regional Weight:** The donut chart shows that while many nations spend, a few Global Regions like the Americas and Asia carry most of the financial weight.
- **Long-Term Outliers:** The United States and China are identified as the consistent primary drivers of the spending curve.

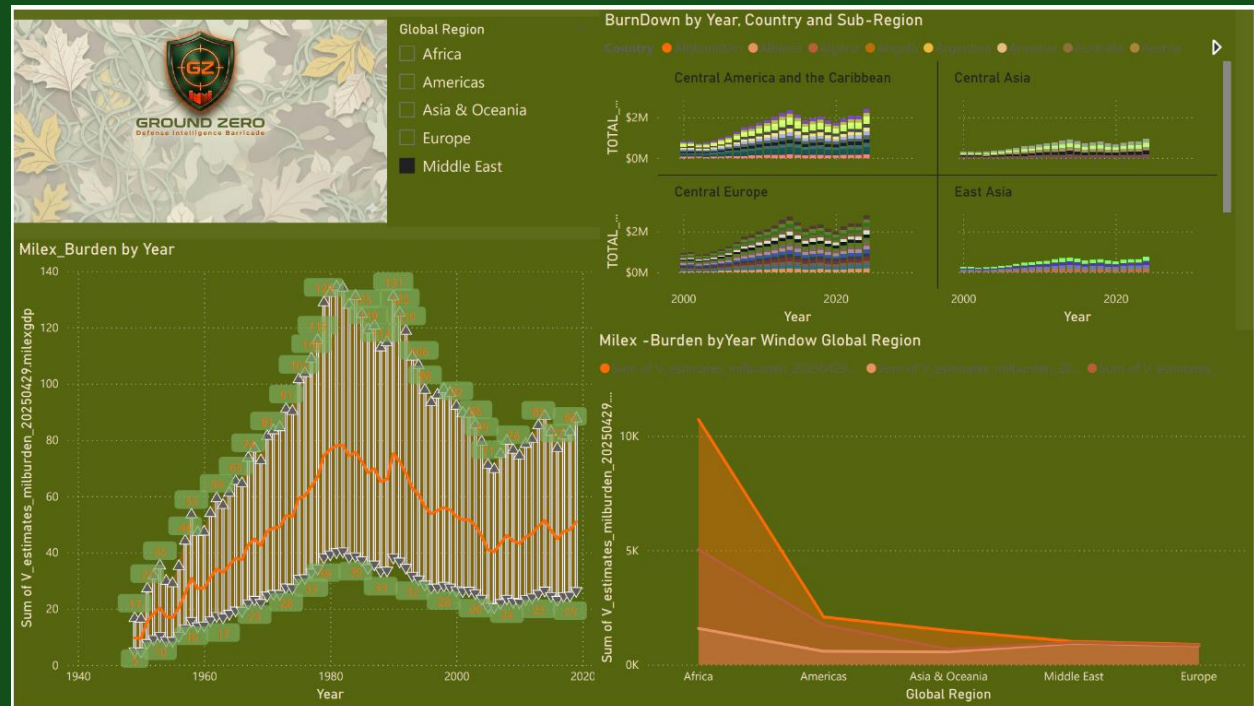




3. Economic Burden & Regional Volatility

GDP Pressure & Regional Burn-Down

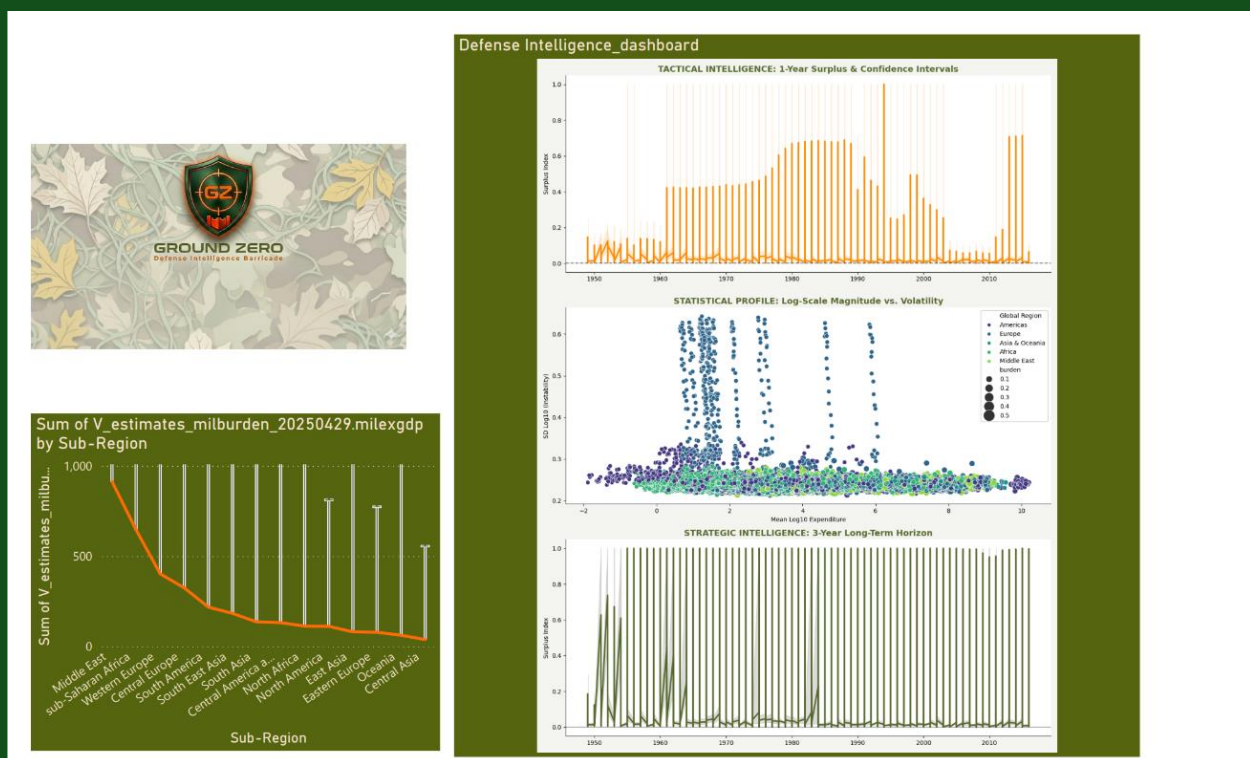
- This view focuses on the "Milex Burden", military spending as a percentage of GDP to show the economic strain on different regions.
- **Economic Pressure:** The main line chart shows how the burden has fluctuated historically, and here a high line indicates periods where defense consumed a larger slice of the economic pie.
- **Regional Horizon:** The small multiples (Central Europe, East Asia, etc.) allow for cross comparisons of spending intensity.
- **The Africa Spike:** The Pareto chart highlights that while Africa's absolute spend may be lower, its regional "Milex Burden" is disproportionately high.



- This advanced analytical view uses Python-driven statistical models to identify surplus spending and statistical anomalies. **Tactical Surplus (Top)**: Shows 1-year windows of spending spikes. This is your "Early Warning System" for sudden military escalations.



- **Volatility Matrix (Middle):** This scatter plot maps expenditure magnitude against instability (SD). It identifies which regions are stable heavy - spenders versus volatile risks.
- **Strategic Horizon (Bottom):** Uses a 3-year smoothed view to identify long-term structural shifts in military posture.





5. Corporate Intelligence Ledger

The Defense Prime Matrix

- This is a tabular "deep dive" into the revenue of the world's most powerful defense contractors by their home country.
- **China's Rise:** The massive, consolidated revenue from the three primary Chinese-owned enterprises (Aviation, North Industries, and Shipbuilding) are quite notable leading next to USA.
- **Western Consolidation:** Compare the aggregated revenue of U.S. giants (Lockheed, Raytheon, Boeing) against European leaders like BAE Systems.
- **Market Value:** Use this to discuss the Total Addressable Market (\$511B+ in this view) controlled by the top-tier primes.



Company Name	China	United Kingdom	United States Of America	Total
Aviation Industry Corporation of China	● \$67,911.42			\$67,911.42
BAE Systems		◆ \$24,739.35		\$24,739.35
Boeing			● \$58,158	\$58,158
China North Industries Group Corporation Limited	● \$70,303.18			\$70,303.18
China State Shipbuilding Corporation Limited	● \$66,911.23			\$66,911.23
General Dynamics			▲ \$37,900	\$37,900
L3Harris Technologies			◆ \$18,194	\$18,194
Lockheed Martin			● \$65,398	\$65,398
Northrop Grumman			▲ \$36,799	\$36,799
Raytheon Technologies			● \$65,000	\$65,000
Total	\$205,125.83	\$24,739.35	\$281,449	\$511,314.18

Conclusion

This report details the final architecture and analytical findings of the Global Military-Industrial Base Analysis project. By synthesizing complex datasets, we have created a unified platform to evaluate the relationship between national fiscal policy and industrial output.

The final analysis reveals a complex interdependency between national wealth and military industrialization.



- **GDP & Defense Dependency:** The data demonstrates that "Industrial Powerhouses" are not defined solely by high GDP, but by their ability to convert national wealth into a self-sustaining domestic defense sector.
- **Regional Shifts:** The longitudinal analysis indicates a clear migration of spending power; while traditional Western powers-maintained dominance in the mid-20th century, the late-century data shows a significant rise in the "Asia & Oceania" region's share of global spending.

Future Work

- **Outlier Detection:** Implementing robust auditing to flag countries whose spending patterns deviate significantly from their historical norms considering other socio-economic growth needs across these countries.
- **Supply Chain Integration:** Incorporating secondary datasets to map the relationship between "Top Manufacturers" and their global subcontractors.
- **Real-Time Fiscal Feeds:** Moving from static CSV downloads to API-based connections for live SIPRI and market updates



Final Reflection

This project proves that military industrialization is an ecosystem. By merging sovereign spending with manufacturer's revenue, we can see that a nation's security is increasingly tied to its industrial capacity. The resulting dashboard provides a predictive lens, showing that current spikes in standardized spending are the primary leading indicators for the next decade's industrial leaders.